The Doppler Effect

> SYNOPSIS
>
> When a source or a listener is moving, the frequency that arrives is not the frequency that left. We work out the two cases separately and then write them as one formula.

BEFORE YOU READ ON — PREDICT

An ambulance is coming towards you with its siren at a steady pitch. **(1)** Is the pitch you hear higher, lower, or the same as the pitch the *driver* hears? **(2)** As it gets closer and closer, does the pitch you hear keep climbing? **(3)** After it has passed, does the pitch keep falling? Write down your three answers. Two of these catch almost everybody.

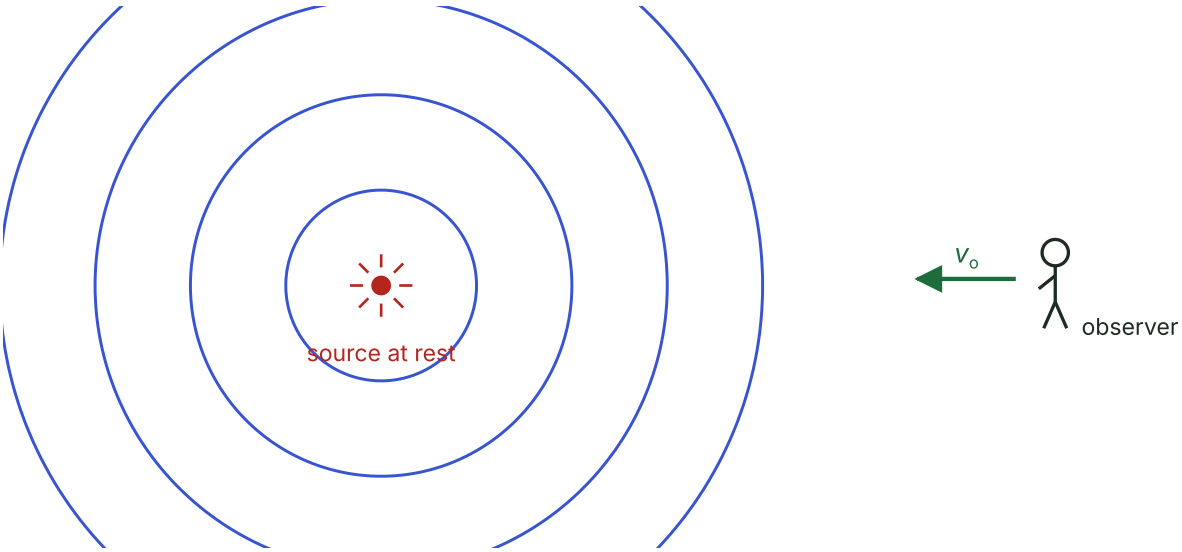
PERFORMANCE EXPECTATIONS

1. **Explain** why an observer moving toward a source meets more wavefronts per second.
2. **Explain** why a moving source crowds the wavefronts ahead of it — a change in wavelength rather than in rate.
3. **Apply** the sign convention to decide, *before* calculating, whether the arriving frequency rises or falls.
4. **Compute** the observed frequency for a moving source, for a moving observer, and for both at once.

Every note in this module so far has had the source and the listener sitting still in the medium. Note 11 let the *intensity* fall off with distance, but the frequency that arrived was always the frequency that left. Now let one of them move.

A moving observer

Start with the easier half: the source stays put and *you* move. Nothing about the wave in space changes — the source is still sitting where it was, sending out a crest every T seconds.



Mark the distance between two crests on this figure and write what it equals. Then write how fast the crests go past the running observer.

1. A source at rest sends out a crest every T seconds. How far apart are the crests in space?
2. Now you run *towards* the source at v_o. How fast do the crests go past you, and so how many do you meet each second?
3. Write f_o. What changes if you run *away* instead?

A moving source

Now the other half, and it is **physically a different thing** even though the formula will come out looking symmetric. This time you stand still and the source moves while it is sounding — so each crest leaves from a different place.



The three grey dots are where the source was one, two and three periods ago. Draw the crest that left from each — each has been spreading at v ever since — then mark the spacing ahead of the source and behind it.

4. In one period the wave travels vT, and the source itself travels v_sT. So how far apart are the crests *ahead* of the source?

5. How fast do those crests travel towards you? Turn the new wavelength into the frequency you hear.

6. And behind the source?

Both at once

7. Put the two results into a single formula, and say what decides each sign.

THE PUNCHLINE

v = speed of sound. **+ : O moving towards. − : S moving towards.**
Notice what is *not* in it: distance. How far away the ambulance is makes no difference to the pitch at all — only how fast it is closing on you. That is the single most useful thing on this page, and it is also the thing almost everyone gets wrong the first time.

Looking ahead

That closes Module 1. You started with a wave on a string in a simulation and a ruler, and you finish holding the equation of motion for a mechanical wave, the condition that picks out a guitar's notes, the energy an oscillation carries, the way that energy thins with distance, and what happens to all of it when things start moving.

Where this sits. Notes 1 to 7 built the wave, Note 8 trapped it, Notes 10 and 11 followed its energy out into the room, and this note moves the furniture. One thread runs through every one of them: **the medium sets the speed.** It set $v = \sqrt{T/\mu}$ in Note 7, it decided which wavelengths fit in Note 8, it carried the intensity outward in Note 11, and here it is again — the crests travel at v whatever the source is doing, which is exactly why a moving source has to change the wavelength instead.

BACK TO YOUR PREDICTION

(1) Higher than the driver hears. The driver is moving with the siren, so for her nothing is approaching anything. **(2) No — it stays constant.** Look at the formula: there is no r in it. At a steady speed towards you the pitch is steady and high, however close it gets. **(3) No — also constant, and low.** The entire change happens *during* the pass, not before or after it. (Strictly, what counts is the part of v_S pointing along the line to you, which sweeps from $+v_S$ to $-v_S$ as it goes by — so the drop is a fast slide rather than a step, and at the instant it is level with you, you hear its true pitch.)